



CHRIST
(DEEMED TO BE UNIVERSITY)
DELHI - NCR, INDIA

Advance Python Programming

MCA-372

CIA – 02

BY

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SUBMITTED TO

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1. Use Excel dataset and draw the graph
Bar chart of Breed vs average Energy_Level
Pie chart of Favorite_Toy distribution
Line graph of Age_Months vs Weight_kg
Print which breed has highest Energy level, most favorite toy, which age has least weight and most weight on the website.
Display them on Website using Streamlit

Ques 1 :

```
import streamlit as st
import pandas as pd
import matplotlib.pyplot as plt

# Title of the app
st.title("🐱 Kitten Data Analysis Dashboard")

# Load the dataset
data = pd.read_csv("./big_kitten_dataset.csv")

# Display the dataset preview
st.subheader("Dataset Preview")
st.write(data.head())

# 1. Bar chart: Breed vs Average Energy Level
st.subheader("Bar Chart: Breed vs Average Energy Level")
avg_energy = data.groupby("Breed")["Energy_Level"].mean()
st.bar_chart(avg_energy)

# 2. Pie chart: Favorite Toy Distribution
st.subheader("Pie Chart: Favorite Toy Distribution")
toy_distribution = data["Favorite_Toy"].value_counts()
fig, ax = plt.subplots()
ax.pie(toy_distribution, labels=toy_distribution.index, autopct="%1.1f%%",
startangle=90)
ax.set_title("Favorite Toy Distribution")
st.pyplot(fig)

# 3. Line graph: Age (Months) vs Weight (kg)
st.subheader("Line Graph: Age (Months) vs Weight (kg)")
data_sorted = data.sort_values("Age_Months")
fig, ax = plt.subplots()
ax.plot(data_sorted["Age_Months"], data_sorted["Weight_kg"], marker="o",
color="green")
ax.set_xlabel("Age (Months)")
ax.set_ylabel("Weight (kg)")
ax.set_title("Age (Months) vs Weight (kg)")
st.pyplot(fig)
```

```
# 4. Insights
st.subheader("Insights")
highest_energy_breed = avg_energy.idxmax()
most_favorite_toy = toy_distribution.idxmax()
min_weight_age = data.loc[data["Weight_kg"].idxmin(), "Age_Months"]
max_weight_age = data.loc[data["Weight_kg"].idxmax(), "Age_Months"]

st.write(f"Breed with the highest average Energy Level:
**{highest_energy_breed}**")
st.write(f"Most favorite toy: **{most_favorite_toy}**")
st.write(f"Age with the least weight: **{min_weight_age} months**")
st.write(f"Age with the most weight: **{max_weight_age} months**")
```

Output :

**You Can Run it by entering the command in the terminal by using
“streamlit run Que1.py”**

Deploy :

Kitten Data Analysis Dashboard

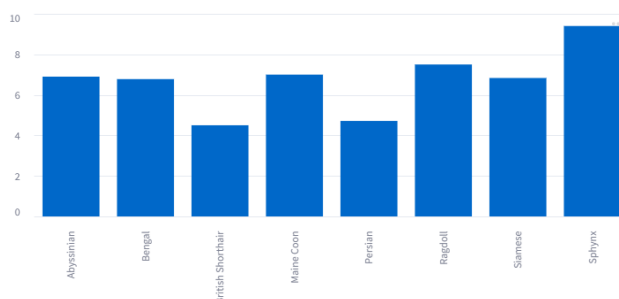
Dataset Preview

	Kitten_ID	Breed	Age_Months	Weight_kg	Favorite_Toy	Energy_Level	Adopted
0	K001	Maine Coon	5	0.9	String	8	No
1	K002	Maine Coon	4	2.1	Mouse	10	Yes
2	K003	British Shorthair	2	1	Mouse	4	No
3	K004	Siamese	8	2.2	Laser Pointer	6	Yes
4	K005	Sphynx	4	1.2	String	10	No

Bar Chart: Breed vs Average Energy Level

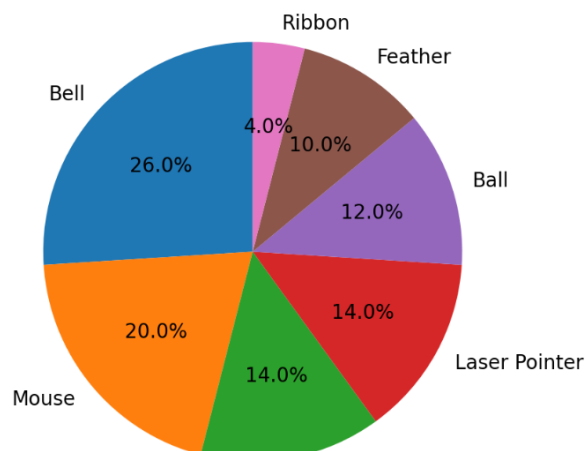


Bar Chart: Breed vs Average Energy Level

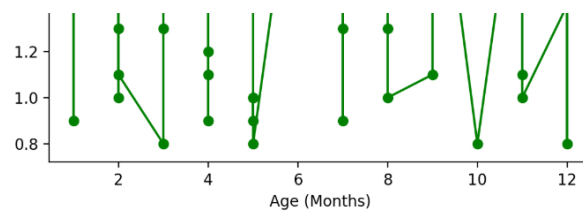
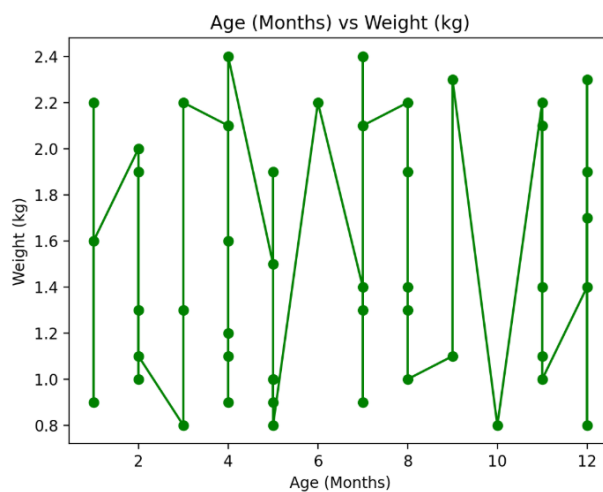


Pie Chart: Favorite Toy Distribution

Favorite Toy Distribution



Line Graph: Age (Months) vs Weight (kg)



Insights

Breed with the highest average Energy Level: **Sphynx**

Most favorite toy: **Bell**

Age with the least weight: **5 months**

Age with the most weight: **4 months**

Ques 2 :

2. Consider the image,
 - 1) Get the details about the image
 - 2) make the image squared image
 - 3) Consider the original image and crop it to get only the face of the kitten.
 - 4) Apply filter
 - 5) Flip and rotate the image, customize
 - 6) Apply enhancement features

Output :

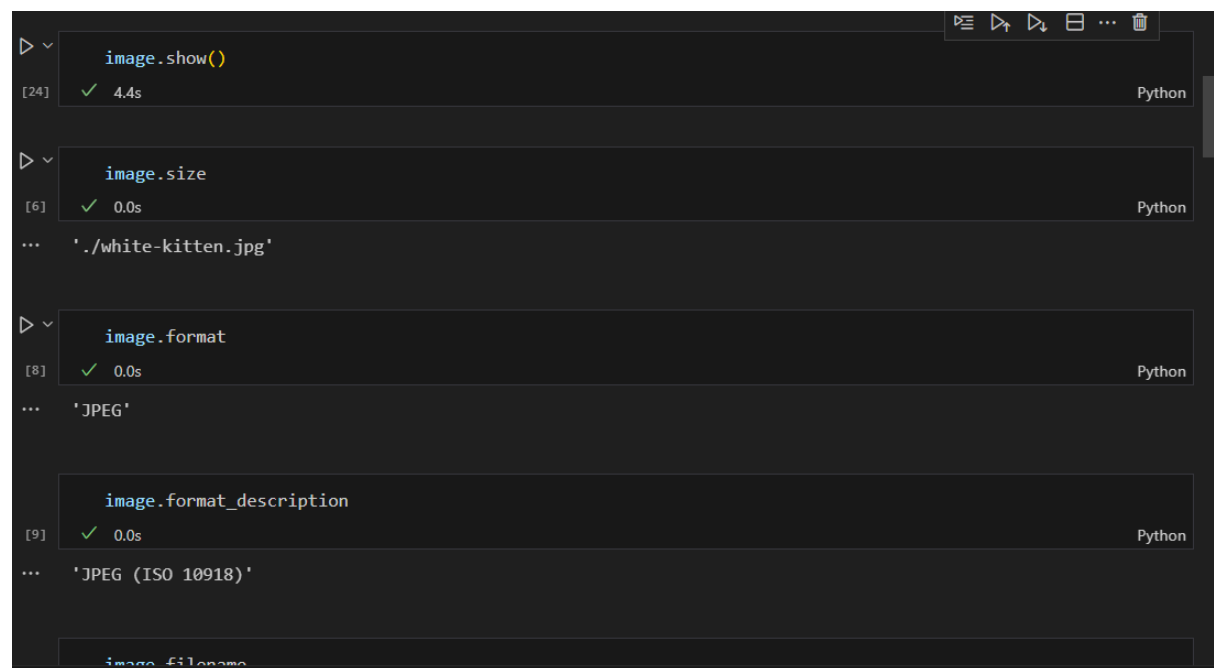
Importing Library :

```
from PIL import Image
```

Importing Image :

```
image = Image.open("./white-kitten.jpg")
```

1. Get the Details about the image



```
[6] ✓ 0.0s Python
... './white-kitten.jpg'

image.size

[8] ✓ 0.0s Python
... 'JPEG'

image.format

[9] ✓ 0.0s Python
... 'JPEG (ISO 10918)'

image.format_description

[7] ✓ 0.0s Python
... './white-kitten.jpg'

image.filename
```

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2. Make the Image Squared image

```
min_dimension = min(image.size)
squared_image = image.crop((0, 0, min_dimension, min_dimension))
squared_image.show()

[10] ✓ 4.9s Python
```



3. Consider the original image and crop it to get only the face of the kitten.

```
crop_box = (200, 50, 400, 250)  
cropped_image = image.crop(crop_box)  
cropped_image.show()
```

[23] ✓ 4.6s

Python



4. Apply filter

Import the Library for ImageFiltering

```
[13] ✓ 0.0s from PIL import ImageFilter Python
```

```
[15] ✓ 4.5s filtered_image = image.filter(ImageFilter.CONTOUR)
filtered_image.show() Python
```

```
[16] ✓ 4.3s blur_image=image.filter(ImageFilter.BLUR)
blur_image.show() Python
```

```
detail_image=image.filter(ImageFilter.DETAIL)
```

```
[17] ✓ 4.3s detail_image=image.filter(ImageFilter.DETAIL)
detail_image.show() Python
```

```
[ ] edge_enhance_image=image.filter(ImageFilter.EDGE_ENHANCE)
edge_enhance_image.show() Python
```

```
[ ] edge_enhancemore_image=image.filter(ImageFilter.EDGE_ENHANCE_MORE)
edge_enhancemore_image.show() Python
```

```
[ ] sharpen_image=image.filter(ImageFilter.SHARPEN)
sharpen_image.show() Python
```

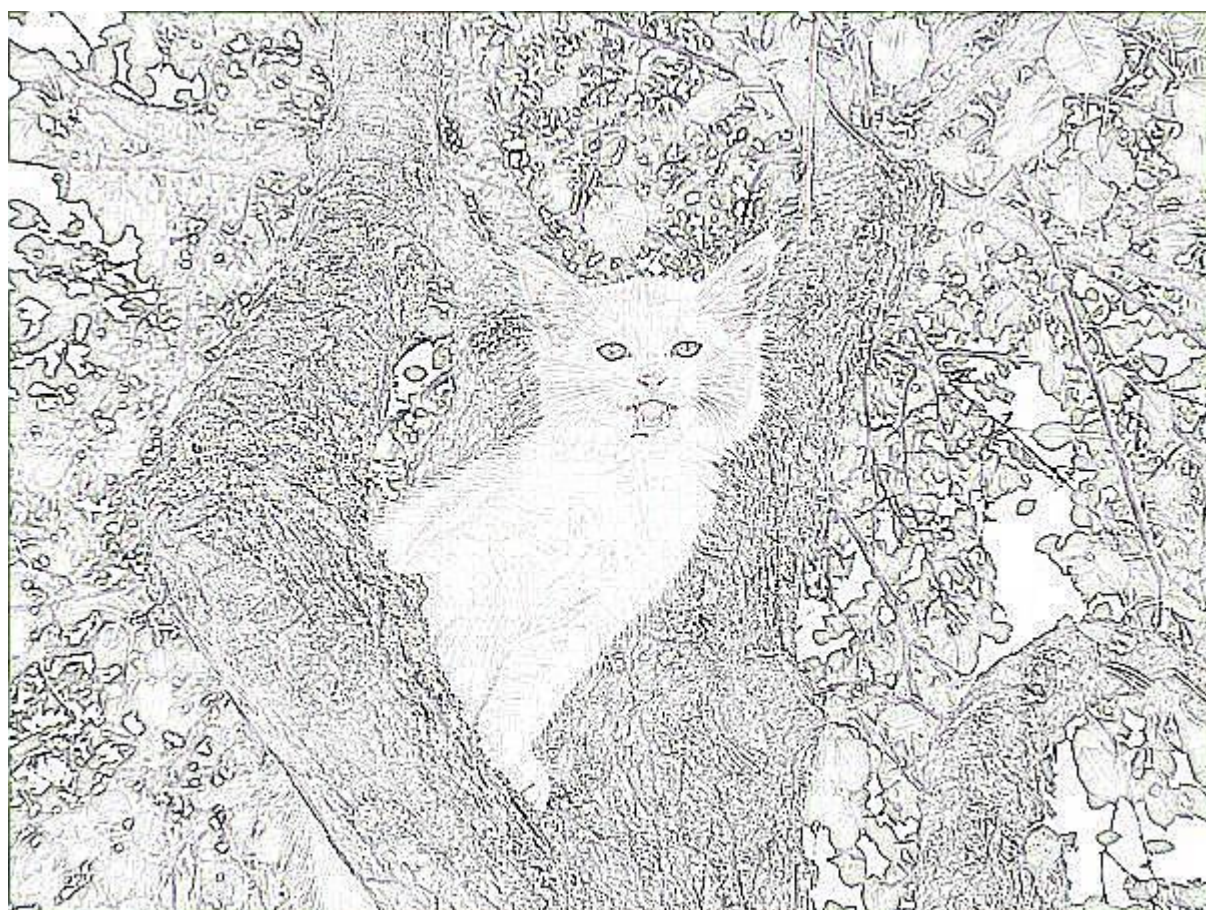
```
[ ] smooth_image=image.filter(ImageFilter.SMOOTH)
smooth_image.show() Python
```

```
[14] ✓ 4.5s smooth_image=image.filter(ImageFilter.SMOOTH_MORE)
smooth_image.show() Python
```

Generate

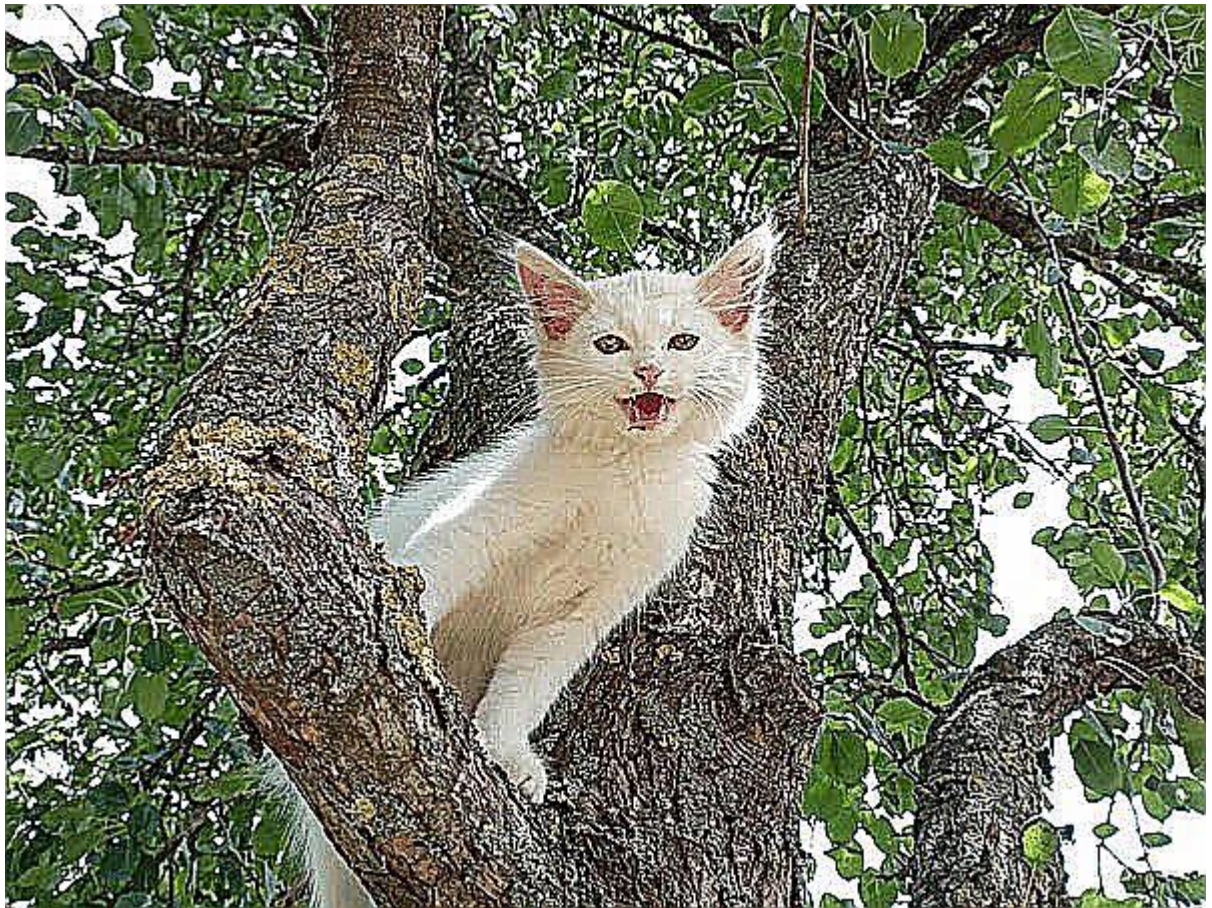
+ Code

+ Markdown











5 Flip and rotate the image, customize

```
image_h=image.transpose(Image.Transpose.FLIP_LEFT_RIGHT)
image_h.show()
```

[18]



4.4s

Python

```
image_h=image.transpose(Image.Transpose.FLIP_TOP_BOTTOM)
image_h.show()
```

[20]

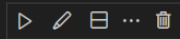


4.5s

Python



6 Apply enhancement features



Importing Libraries

Generate

+ Code

+ Markdown

Start Chat to Generate Code (Ctrl+K I)

```
[21] ✓ 0.0s Python
```

```
from PIL import ImageEnhance
```

```
[22] ✓ 4.5s Python
```

```
image_col=ImageEnhance.Color(image)
colored_image=image_col.enhance(1)
colored_image.show()
```

```
[ ] Python
```

```
image_con=ImageEnhance.Contrast(image)
contrasted_image=image_con.enhance(1.20)
contrasted_image.show()
```

```
[ ] Python
```

```
image_sharp=ImageEnhance.Sharpness(image)
sharped_image=image_sharp.enhance(500)
sharped_image.show()
```

```
[ ] Python
```

```
image_bright=ImageEnhance.Brightness(image)
brightened_image=image_bright.enhance(1)
brightened_image.show()
```

